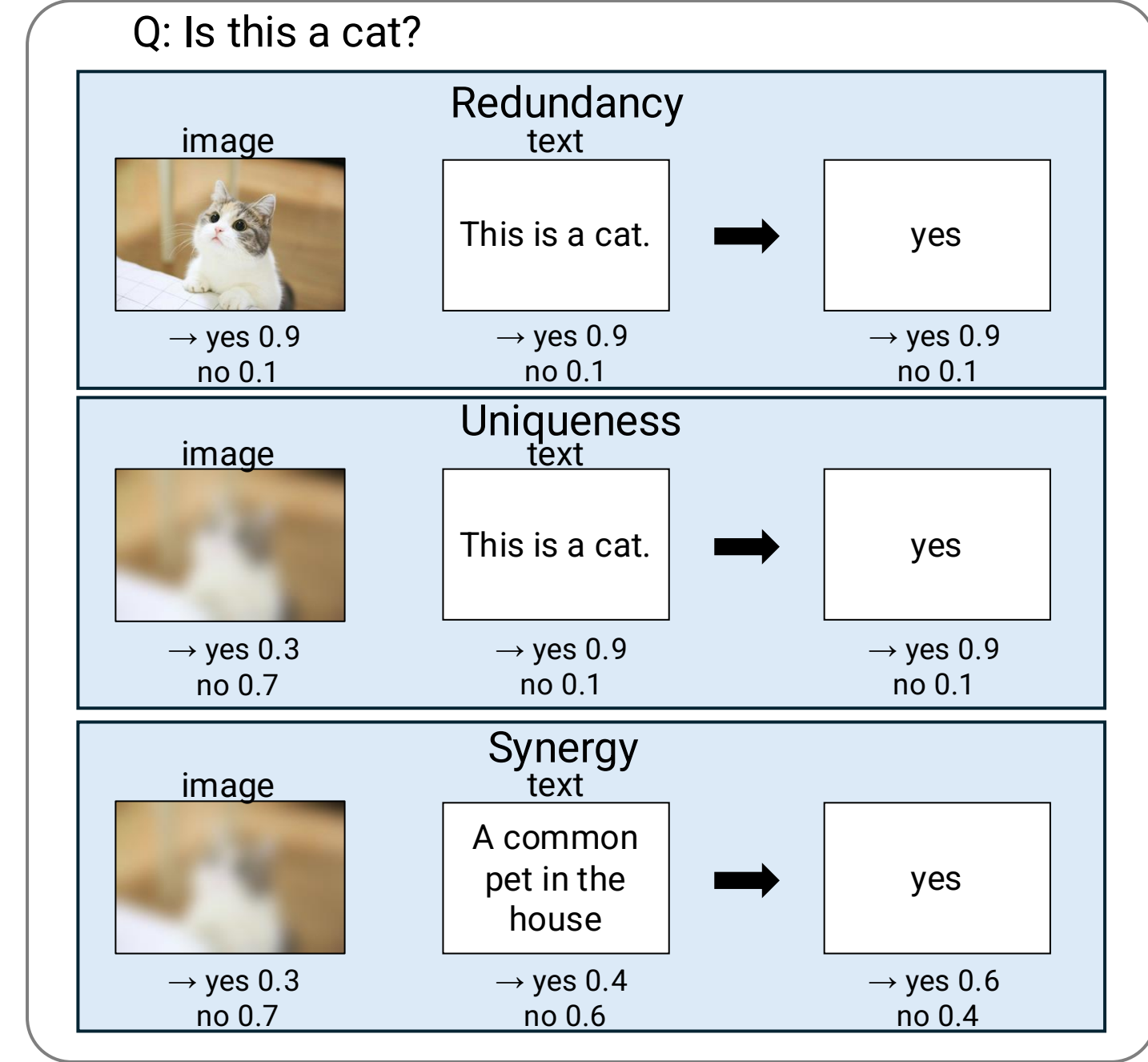


1. INTRODUCTION

Motivation / Problem

- Existing MXAI shows where the model looks, but not how modalities interact.
- Plausible saliency can hide reliance on a single modality.
- Current interaction methods rarely expose fine-grained feature-level evidence.

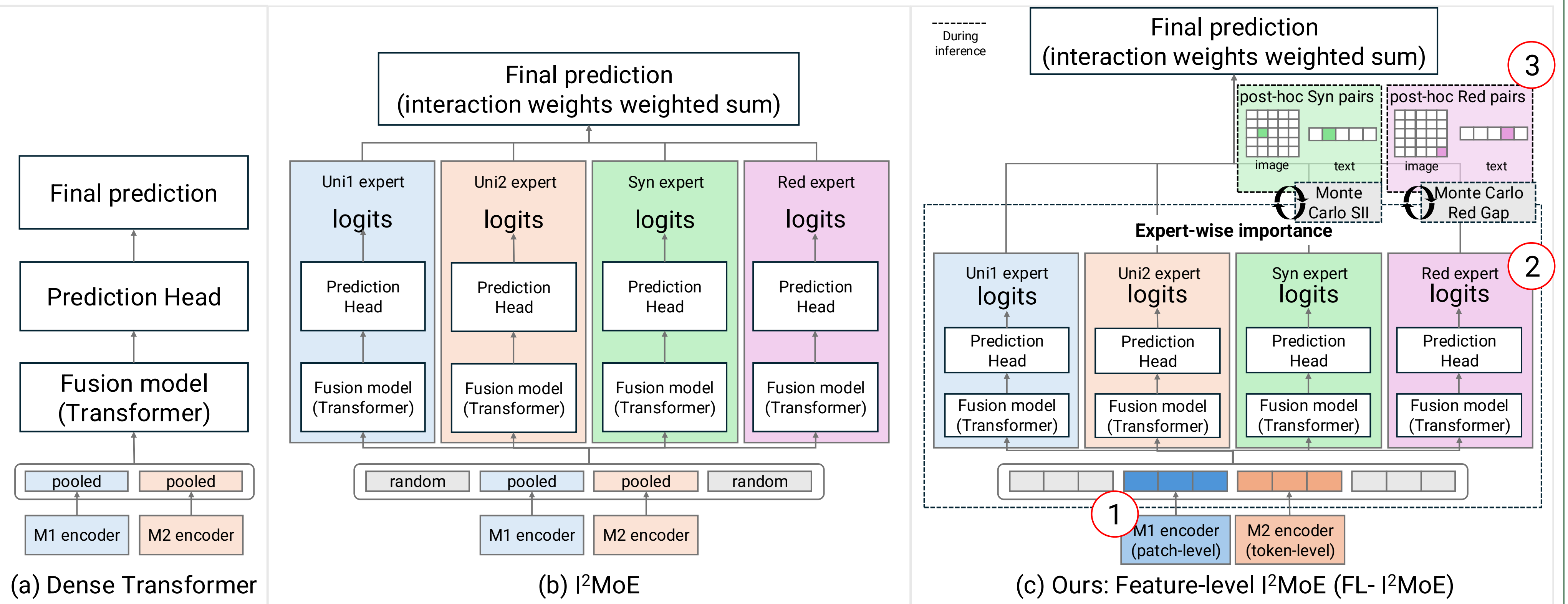
Goal: explain multimodal predictions by separating unique, synergistic, and redundant evidence at the feature level.



Contributions

- We introduce Feature-level I²MoE (FL-I²MoE), which preserves token/patch sequences from frozen pretrained encoders and separates unique, synergistic, and redundant evidence.
- We develop an expert-wise explanation pipeline and evaluate its faithfulness using top-K% masking.
- We use Monte Carlo SII and redundancy-gap scores with pair-level masking to identify and validate concrete cross-modal interactions.

2. METHOD OVERVIEW



1 Feature-level architecture

- Original I²MoE compresses each modality into one pooled representation.
- Our FL-I²MoE replaces pooled encoders with feature-level encoders (ViT for images and RoBERTa for text)

2 Expert-wise Attribution & Candidate Selection

- Compute expert-wise GradXAttnRoll maps for the target-class score.
- Select the top-p valid features per modality and form cross-modal candidate pairs.

3 Inference-time interaction analysis

Estimate SII for synergy and the redundancy-gap for substitutable evidence

- Synergy

$$\Delta_{uv}(S) = f(S \cup \{u, v\}) - f(S \cup \{u\}) - f(S \cup \{v\}) + f(S)$$

$$SII(u, v) = \mathbb{E}_s[\Delta_{uv}(S)]$$

- Redundancy

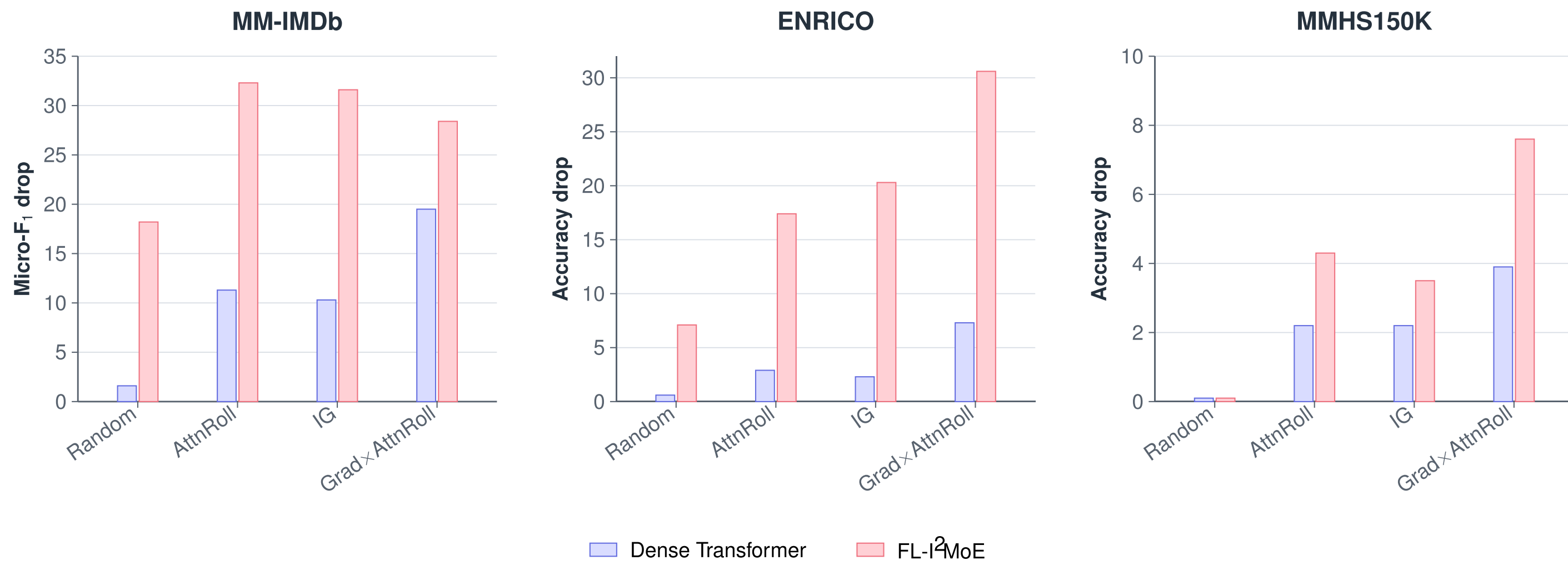
$$g_u = f(S + u) - f(S), g_v = f(S + v) - f(S), g_{uv} = f(S + u + v) - f(S), g_{max} = \max(g_u, g_v)$$

$$base = \max(0, \min(g_{uv}, g_{max})), span(S) = |g_{uv}(S) - g_{max}(S)|$$

$$R_{red}(u, v) = \frac{\mathbb{E}_s[base]}{1 + \mathbb{E}_s[span]}$$

3. KEY RESULTS

3a. Faithfulness of Expert-wise Importance



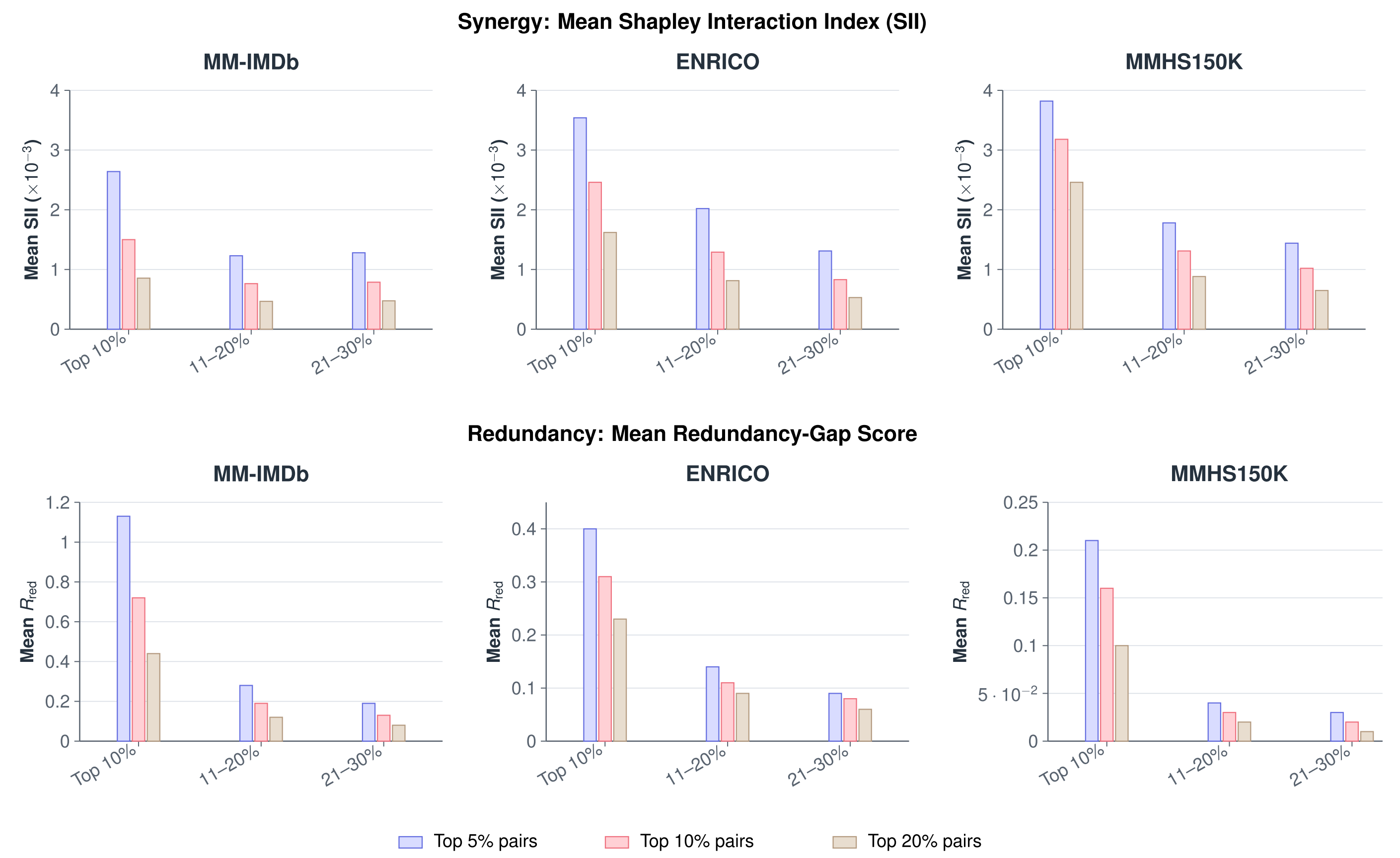
- Attribution-based masking causes larger drops than random masking for both models. FL-I²MoE shows larger drops than the dense Transformer, consistent with more concentrated and faithful importance maps

3c. Pair-level Masking at K=5%

	MM-IMDb		ENRICO		MMHS150K	
Masking setting	Synergy	Redundancy	Synergy	Redundancy	Synergy	Redundancy
Original	0.6789	0.6789	0.5479	0.5479	0.7004	0.7004
Random pairs	0.5703	0.6136	0.5137	0.5240	0.6796	0.6839
Interaction pairs	0.5600	0.5932	0.4795	0.5171	0.6708	0.6745

- At K=5%, masking interaction-ranked pairs from the same top-10% feature pool consistently causes larger performance drops than masking random pairs

3b. Alignment with Monte Carlo Feature Interactions



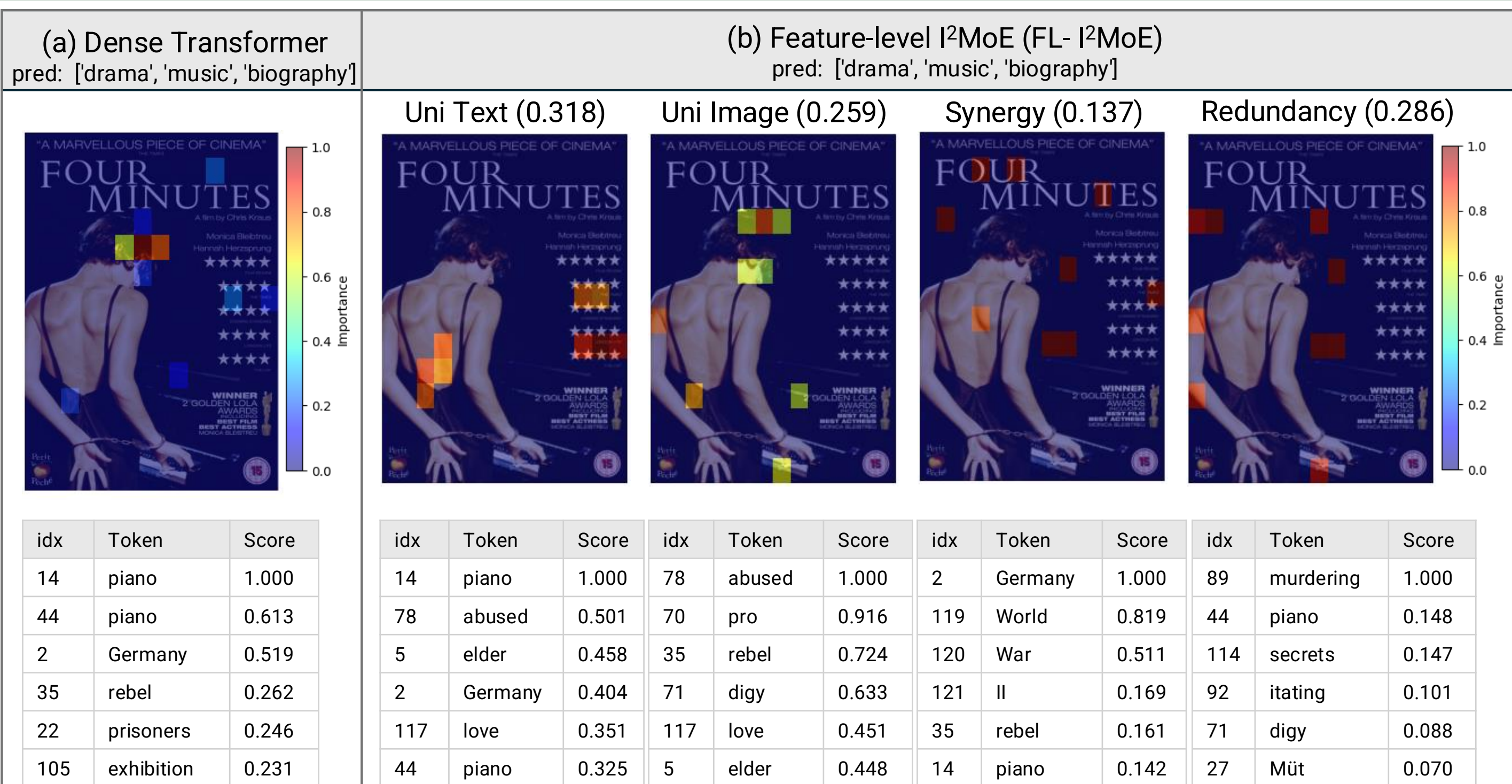
- Across all three datasets, higher-importance feature bins consistently yield larger positive SII and larger redundancy-gap scores

3d. Ablation: Effect of Feature-level Encoders

	MM-IMDb		ENRICO	
Metric	Original I ² MoE	FL-I ² MoE	Original I ² MoE	FL-I ² MoE
Micro-F1	60.33 ± 0.47	67.85 ± 0.45	-	-
Macro-F1	51.21 ± 0.57	56.35 ± 0.15	-	-
Accuracy	-	-	45.43 ± 1.13	51.26 ± 2.17

- Feature-level encoders improve task performance while enabling structured interpretability

4. QUALITATIVE ANALYSIS



5. CONCLUSION

- FL-I²MoE explains multimodal predictions at the feature-pair level, separating unique, synergistic, and redundant evidence.
- Across three benchmarks, FL-I²MoE produces more concentrated and faithful expert-wise explanations than a dense Transformer.
- SII and redundancy-gap scores identify cross-modal pairs that are causally relevant under pair-level masking.